stats101a_hw5

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```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(ggplot2)
```

(1)

(a)

```r
df <- data.frame(read.table('./indicators.txt', header= TRUE))

str(df)
```

```
## 'data.frame':    18 obs. of  3 variables:
##  $ MetroArea          : chr  "Atlanta" "Boston" "Chicago" "Dallas" ...
##  $ PriceChange        : num  1.2 -3.4 -0.9 0.8 -0.7 -9.7 -6.1 -4.8 -6.4 -3.4 ...
##  $ LoanPaymentsOverdue: num  4.55 3.31 2.99 4.26 3.56 4.71 4.9 3.05 5.63 3.01 ...
```

```r
sapply(df, anyNA)
```

```
##           MetroArea         PriceChange LoanPaymentsOverdue
##               FALSE               FALSE               FALSE
```

```r
model <- lm(PriceChange ~ LoanPaymentsOverdue, data = df)
confint(model, level = 0.95)
```

```
##                          2.5 %     97.5 %
## (Intercept)          -2.532112 11.5611000
## LoanPaymentsOverdue -4.163454 -0.3335853
```

A 95% confidence interval for the slope of the regression model (β_1) is (-4.163454, -0.3335853).

There is evidence of a significant negative linear association between `PriceChange` and `LoanPaymentsOverdue` at the 5% significane level. The interval lies entirely below 0, so the slope is significanty less than 0.

```r
predict(model, newdata = data.frame(LoanPaymentsOverdue = 4))
```

```
##         1
## -4.479585
```

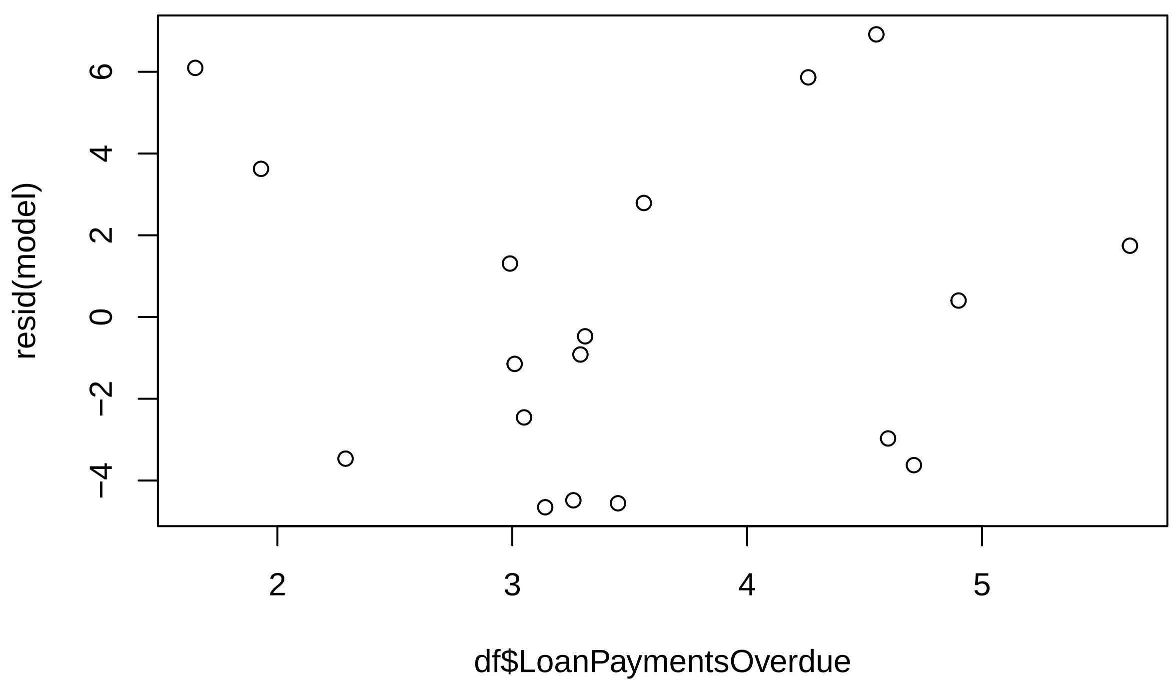
```r
predict(model, newdata = data.frame(LoanPaymentsOverdue = 4), interval = "confidence", level = 0.95)
```

```
##         fit       lwr       upr
## 1 -4.479585 -6.648849 -2.310322
```

No. 0% is not a feasiable value. It is not inside the interval (-6.648849, -2.310322)

(b)

```r
plot(df$LoanPaymentsOverdue, resid(model))
```



Seems to be a bit gray. There's no strong trend, but the residuals show some clustering and uneven spread. (maybe non-linearity)?

the variance also does not look the most constant (some regions are more tightly clustered)

it's not completely valid nor invalid. Somewhere in the middle I think.

(2)

```r
df <- data.frame(read.table('./invoices.txt' ,header = TRUE))

str(df)
```

```
## 'data.frame':    30 obs. of  3 variables:
##  $ Day     : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Invoices: int  149 60 188 23 201 58 77 222 181 30 ...
##  $ Time    : num  2.1 1.8 2.3 0.8 2.7 1 1.7 3.1 2.8 1 ...
```

(a) Find a 95% confidence interval for the start-up time, i.e., b0.

```r
model <- lm(Time ~ Invoices, data = df)

confint(model, level = 0.95)
```

```
##                   2.5 %     97.5 %
## (Intercept) 0.391249620 0.89217014
## Invoices    0.009615224 0.01296806
```

(0.391249620, 0.89217014)

Suppose that a best practice benchmark for the average processing time for an additional invoice is 0.01 hours (or 0.6 minutes). Test the null hypothesis H b= against a two-sided alternative. Interpret your result.

```r
b1 <- coef(model)["Invoices"]                     # estimated slope
se_b1 <- summary(model)$coefficients["Invoices", "Std. Error"]

t <- (b1 - 0.01) / se_b1

p <- 2 * pt(-abs(t), df.residual(model))

p
```

```
##  Invoices
## 0.1257402
```

Since p value = 0.13 > 0.05, we fail to reject the null hypothesis that $\beta_1 = 0.01$, which means we do not have enough evidence to conclude that $\beta_1 \neq 0.01$.

(c) Find a point estimate and a 95% prediction interval for the time taken to process 130 invoices.

```r
predict(model, newdata = data.frame(Invoices = 130), interval = 'prediction', level = 0.95)
```

```
##        fit      lwr    upr
## 1 2.109624 1.422947 2.7963
```

point estimate = 2.109624 hours 95% prediction interval = (1.422947, 2.7963) hours

(3)

Model 1 shows a strong linear relationship and the points lie close to the regression line. So its residuals are smaller. Model 2 shows little to no linear relatinoship and so the residuals are larger.

Model 1 has a lower RSS than Model 2. This is because Model 1's points are tightly clustered around the line.

Since SYY = SSreg + RSS, and since both models use the same Y (SYY is the same for both), Model 1 has larger SSreg.

Model 1 has smaller RSS so it must have larger SSreg than Model 2.

Therefore the only answer choice that is true is (d), RSS for model 1 is less than RSS for model 2 while SSreg for model 1 is greater htan SSreg for model 2.

(4)

(a)

$RSE = \sqrt{\frac{RSS}{df}}$ so $RSS = RSE^2 * df = (2.418)^2 * 33 = 192.9$

(b)

$R^2 = \frac{SSreg}{SYY}$ and $SYY = SSreg + RSS$ so $SSreg = \frac{R^2}{1-R^2} * RSS = \frac{0.7254}{1-0.7254} * 192.4 = 509.7$

(c)

$MSreg = \frac{SSreg}{df_{reg}} = 509.7/1 = 509.7$

(d)

$SYY = SSreg + RSS = 509.7 + 192.9 = 702.6$

(e)

$r = \sqrt{R^2} = \sqrt{0.7254} = 0.852$

(5)

(a)

```r
mymodel <- function(random.seed, beta_0, beta_1, sigma, x){
  set.seed(random.seed)
  n <- length(x)
  eps <- rnorm(n, mean = 0, sd = sigma)

  y <- beta_0 + beta_1 * x + eps
}
```

Since $Var(\hat{\beta}_1) = \frac{\sigma^2}{SXX} \approx \frac{RSS}{n-2} * \frac{1}{SXX}$, try to maximize SXX.

So have as much variation in x as possible (as far away from mean as possible). So concentrate in extremes.

```r
x <- rep(c(0, 100), each= 50)

y <- mymodel(123, 1, 10, 20, x)

model <- lm(y ~ x)
```

```r
summary(model)
```

```
##
## Call:
## lm(formula = y ~ x)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -49.11 -11.83  -1.16  13.15  42.69
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.68807    2.59003   0.652    0.516
## x           10.02240    0.03663 273.622   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.31 on 98 degrees of freedom
## Multiple R-squared:  0.9987, Adjusted R-squared:  0.9987
## F-statistic: 7.487e+04 on 1 and 98 DF,  p-value: < 2.2e-16
```

The standard error is 0.03663!

(b)

I would explain the standard error of the slope to a beginning statistics student by saying:

it is how much the estimated slope would vary if you repeated the study many times. Fitting a line and getting a slope won't be exact, just like sample point estimates. It depends on your particular sample. So if you collect a different sample and fit the line again, you would get a slightly different slope. So, the standard error of the slope measure show much these slopes differ from one another.